

GRAPH THEORY: NETWORKS & SHORTEST PATH

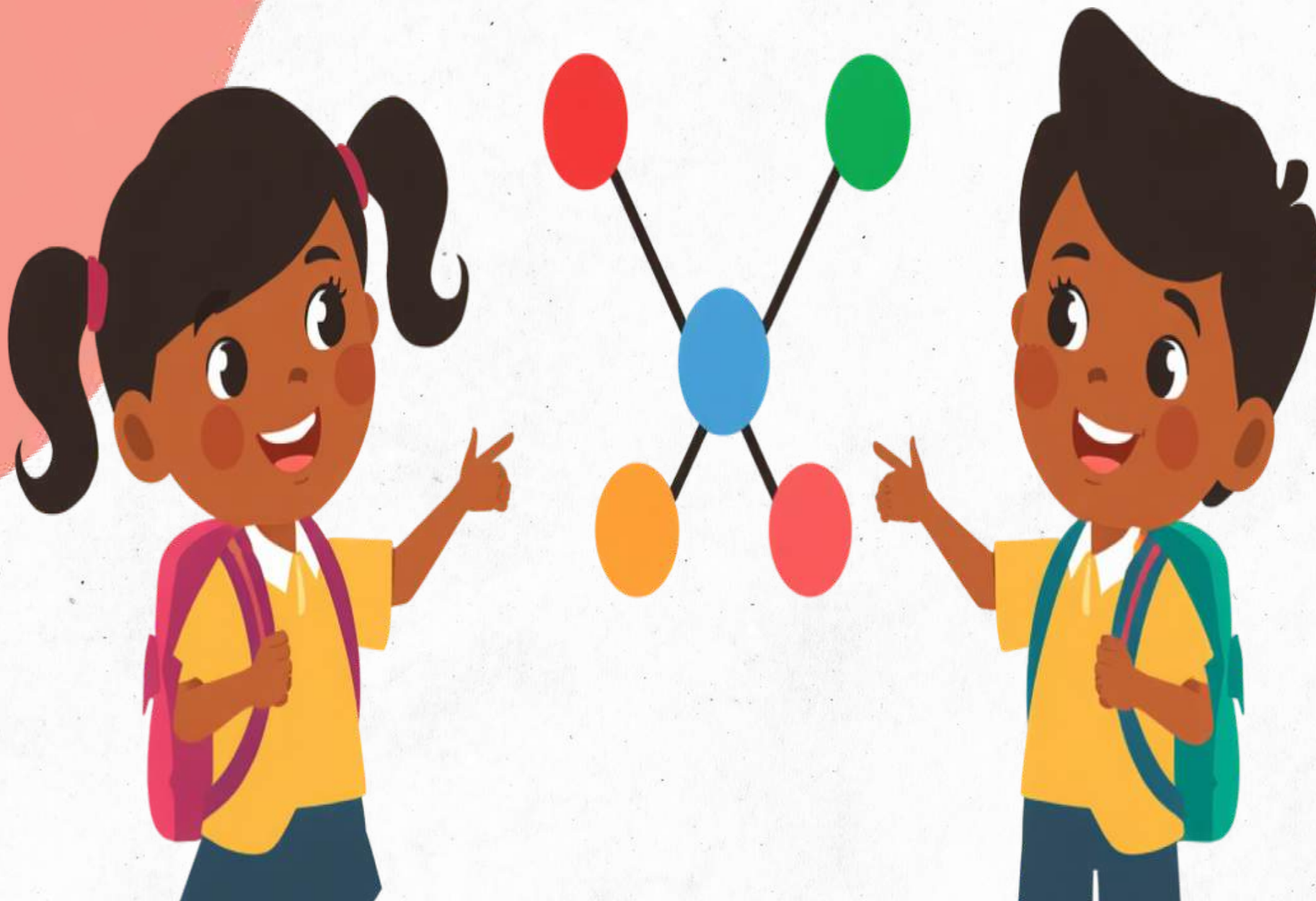


THE GREAT INDIAN WEDDING EXPRESS!

A grand wedding is happening in Delhi! The Wedding Express train must rush guests to the right stations — on time! But wait... which route is the fastest? Let's find out!

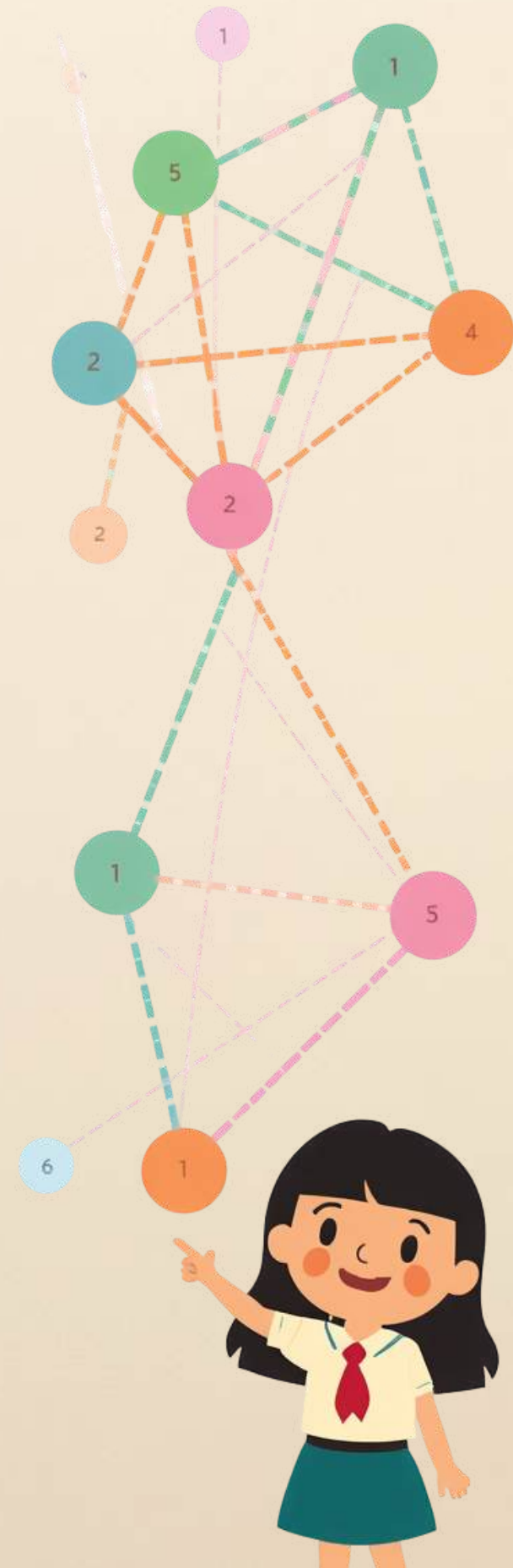
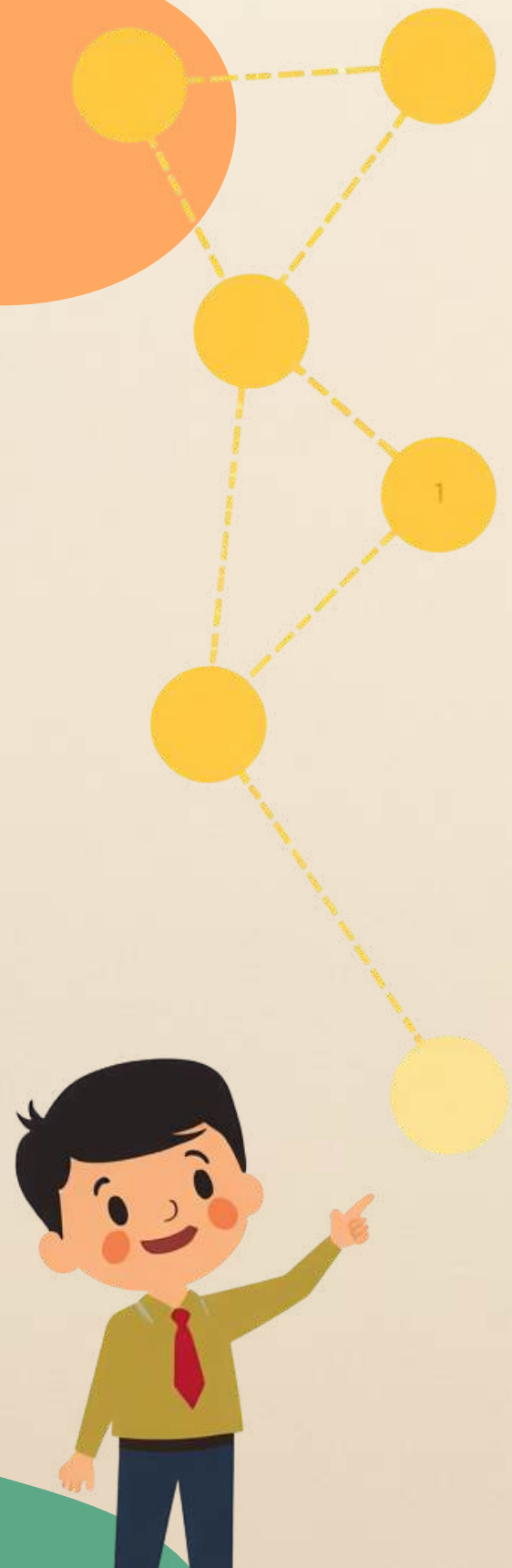


WHAT ARE NODES AND EDGES?



- NODE = A point or place
(like a station or a market!)
 - EDGE = A line or path
connecting two nodes
(like a railway track or a road!)
- Together, nodes + edges
make a NETWORK! 🌐

UNWEIGHTED VS WEIGHTED NETWORKS



Unweighted Network
Edges just show connection
— no numbers!
(e.g., Delhi Metro lines on a
simple map)

Weighted Network
Edges have numbers —
time, cost or distance!
(e.g., 4 min from Rajiv
Chowk to Patel Chowk)

No numbers on
paths — just lines
linking places. All
paths look equal!

Numbers on paths
help us pick the
QUICKEST route!

LET'S EXPLORE THE DELHI METRO MAP!

Look at the mini Delhi Metro map! 🚇

● Stations = Nodes (places to stop)

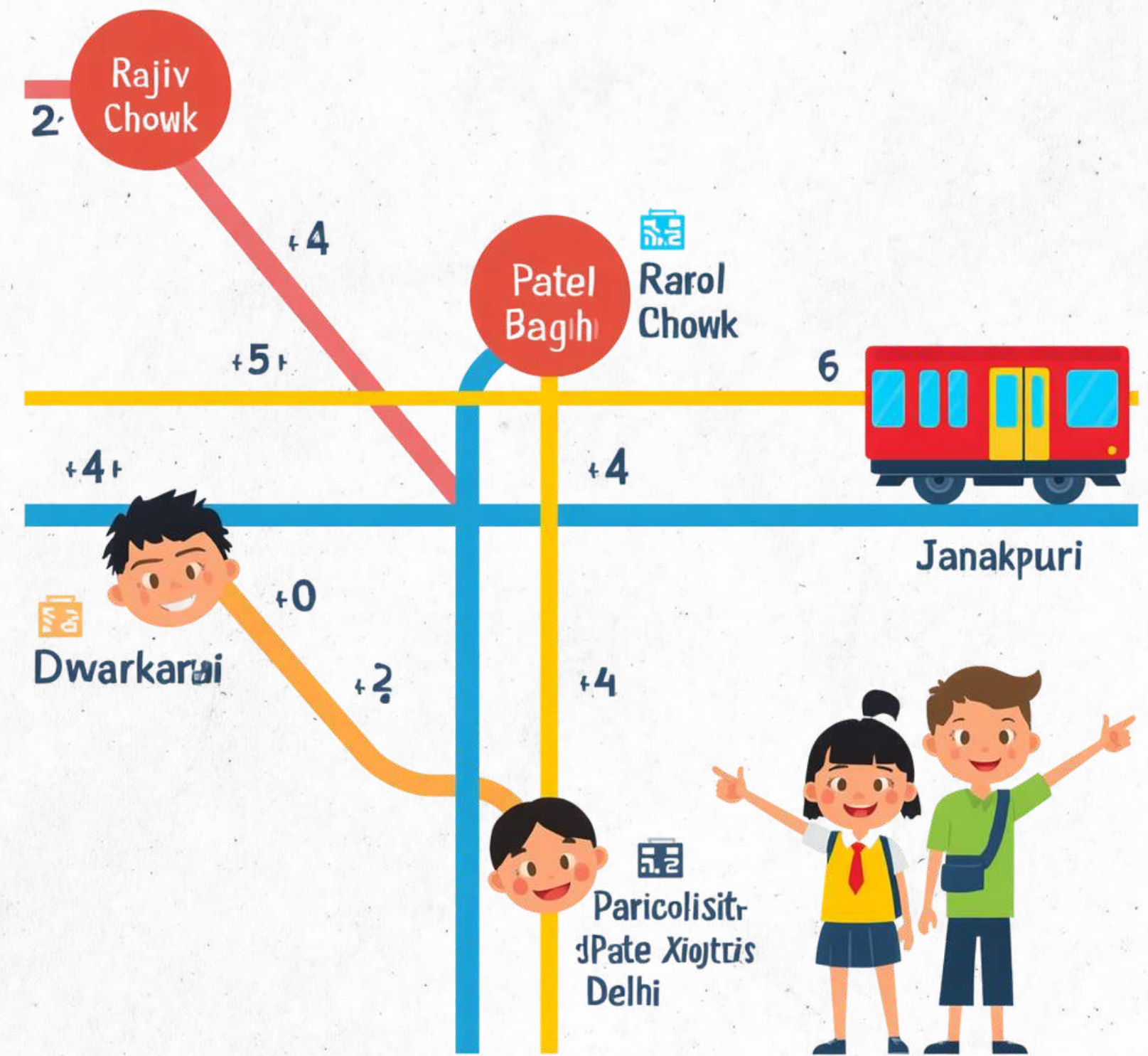
➡ Tracks = Edges (paths connecting them)

🕒 Numbers on tracks = Travel time in minutes

Can you spot the stations?

What connects them?

Which path looks shortest?





MUMBAI DABBAWALA ROUTE CHALLENGE!

The Dabbawalas deliver lunch boxes across Mumbai every day!

Here is a delivery network with 5 stops: A, B, C, D, and E.

Edges (paths) and travel times:

$A \rightarrow B$: 4 min | $A \rightarrow C$: 2 min

$B \rightarrow D$: 5 min | $B \rightarrow E$: 8 min

$C \rightarrow D$: 3 min | $D \rightarrow E$: 2 min

$C \rightarrow E$: 9 min

 Challenge: Find the quickest route from A to E. Add up the times on each path. Which is fastest?

FINDING THE SHORTEST PATH!

1

Step 1: List all possible routes.

$A \rightarrow B \rightarrow E$

$A \rightarrow C \rightarrow D \rightarrow E$

$A \rightarrow B \rightarrow D \rightarrow E$

2

Step 2: Add up the travel times for each route.

3

Step 3: Compare the totals – the smallest sum is your shortest path!

YOUR TURN!

School Bus Route Optimiser 

Stops: Home (A), Park (B), Market (C), School (D)

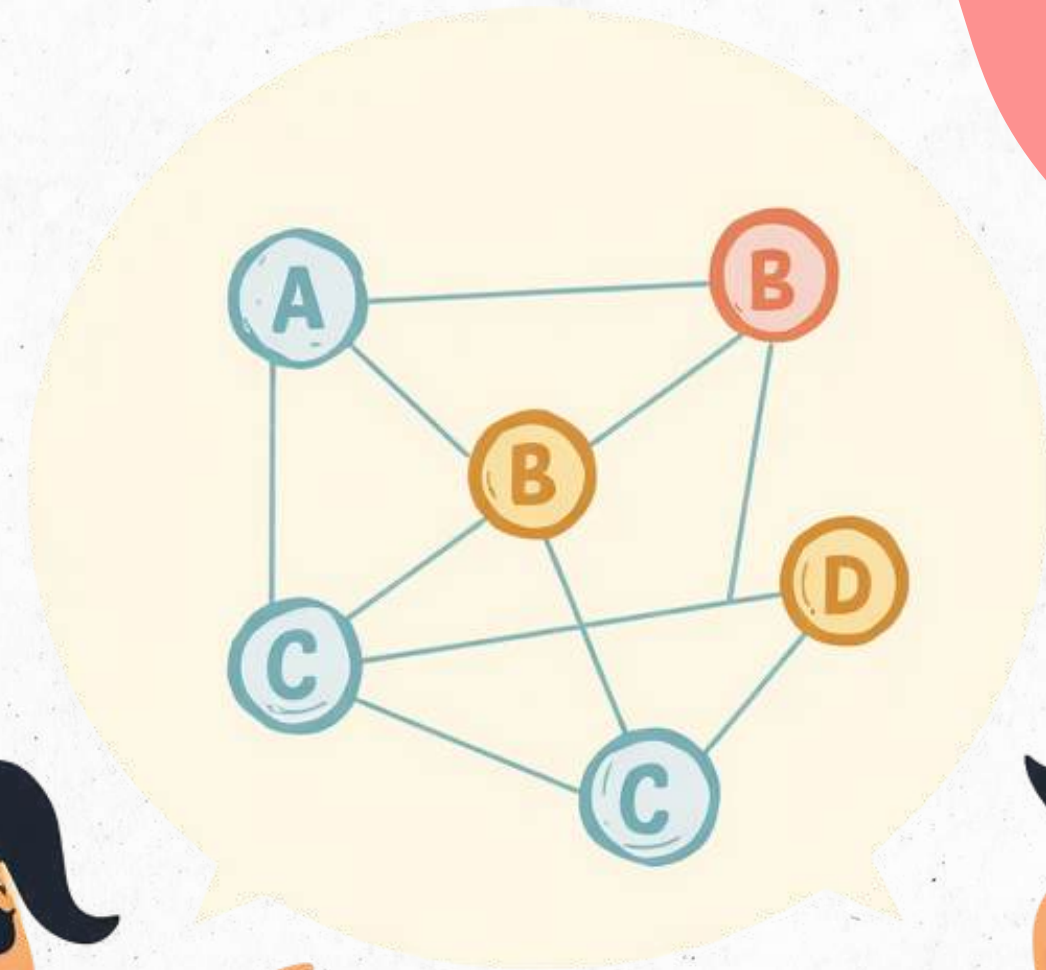
Travel times:

A → B: 4 min | A → C: 7 min

B → D: 5 min | C → D: 3 min

B → C: 2 min

Challenge: Which path from A to D is the quickest? Add up the times and find out!



REAL-WORLD INDIAN ROUTES

Jaipur Auto-
Rickshaw
Routes

Bengaluru
Delivery
Paths

Indian
Railways
Network

Delhi
Metro
Lines

Golden
Quadrilateral
Highway

Local
Bazaar
Connections

Mumbai
Dabbawala
Routes

Traffic vs
Bypass
Times

City
Road
Planning

Highway &
Market
Networks

KEY TAKEAWAYS

1

Nodes = places;
edges = paths
between places.

2

Weighted edges
show travel time
or cost. Shortest
path = quickest
route!

3

Real-world networks
help us plan better
routes every day.
Keep exploring!

HOME CHALLENGE & THANK YOU!

Draw your own network • Label nodes & edges • Find the shortest path!

